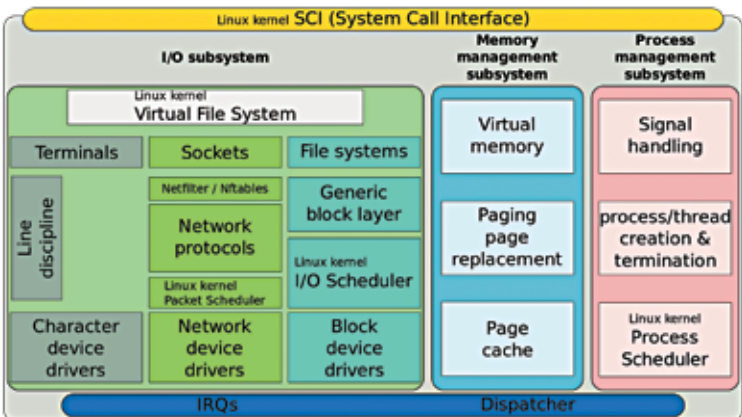


Kernel's architecture has direct impact on the possibilities of the OS. The Linux kernel was conceived in 1991 by Linus Torvalds. His aim was to create an OS which supports all platforms. The Linux kernel is a monolithic kernel. It supports true preemptive multitasking. This type of multitasking involves the use of interrupt mechanism, prioritisation of certain processes and use of memory resources in the most efficient manner.

The Linux kernel is written in the C programming language, and the application programming interface (API) is stable, and can take a lot of load. It's not a stretch to call the Linux kernel the biggest thing to ever happen in open source software! The kernel is developed by contributors world-wide via discussions that take place on forums and the Linux kernel mailing list (LKML).

Android's variant of the Linux kernel has some changes that are implemented by Google outside of the typical Linux kernel development cycle. In contrast to desktop Linux distributions, Android does not give users the ability to access the root to the operating system by default. However, this system can be hacked into and is used by the open source community to improve or enhance the capabilities of the OS. The XDA forum is a good example of this. On top of the Linux Kernel, there are libraries and middleware. Android source code, even though it is on C, does include Java-compatible libraries as well. These libraries were developed by Google exclusively for Android. Google announces major incremental upgrades for Android on a yearly basis based on Linux updates as well as its own updates.



A simplified view of a standard Linux kernel architecture